

MUHAMMAD SAAD

AI Engineer & Full Stack Developer | Production ERP / MES Systems | Python · FastAPI · Claude API

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PROFESSIONAL SUMMARY

AI and full-stack engineer who builds production software that businesses run on daily. Designed, built and deployed a 20+ module enterprise MES/ERP platform now in daily use across production, quality control, finance and procurement at a Dubai manufacturing facility — bridging Sage Evolution ERP with a self-hosted web application built in Python, FastAPI, MongoDB and SQL Server.

Ships production AI systems with the Claude API, OpenAI API and LangChain, including a customer-support copilot with RAG retrieval and citations. Also runs the underlying infrastructure (VMware, Linux, Docker, Nginx, Cloudflare) and works hands-on with automated Krones production lines — a combination of software capability and genuine shop-floor domain knowledge that makes the systems fit how the plant actually operates.

PROFESSIONAL EXPERIENCE

ERP Developer and AI Integration Engineer

Jul 2025 – Present

Kingsley Beverage FZCO | Dubai, UAE · On-site · Full-time

- Designed, built and deployed a 20+ module enterprise MES/ERP platform from scratch, now in daily production use across production, quality control, finance and procurement — replacing manual and Excel-based workflows with a single source of truth across five departments
- Bridged Sage Evolution ERP with a modern self-hosted web application: live inventory synced from SQL Server with automatic offline fallback, customer PO → job orders → batch production reports with OEE tracking → dispatch, and automated GRN posting back into Sage
- Architected a two-database design — SQL Server as the ERP source of truth for stock, MongoDB for application state — with a background sync engine and a dedicated posting microservice for safe writes back into Sage
- Built a full QC documentation suite (batch records, seam checks, mixing sheets, CIP logs) with auto-generated PDF reports, plus FIFO/FEFO lot traceability from goods receipt through to dispatch
- Integrated Claude API and OpenAI API into production workflows for intelligent document processing and automation
- Implemented role-based access (JWT) for admin, managers, QC, mixing area and viewers; reduced production reporting time by an estimated 60% and improved real-time data visibility plant-wide
- Manage the full IT infrastructure: VMware self-hosted server, NAS with RAID storage, Dell server administration, Linux, Docker, Nginx, Cloudflare and facility network architecture
- Act as Production Assistant and Line In-Charge on automated Krones lines (blow molding, filling, Checkmate inspection, Variopac FS packaging, palletizing, depalletizing, PET preform), performing electrical troubleshooting, RCA and preventive maintenance — the domain knowledge the software is built on

NOC Engineer

Jul 2023 – Jul 2025

PTCL | Rawalpindi, Pakistan · On-site · Full-time

- Built and deployed a Python automation tool into the live 24/7 production environment that auto-generated network configuration scripts from incoming service tickets, eliminating manual Excel preparation and significantly reducing provisioning time and configuration errors
- Progressed from L1 Support Engineer to NOC Engineer at Pakistan's national Tier-1 telecom operator
- Monitored GPON, PSTN and broadband network infrastructure using SolarWinds, tracking performance KPIs and proactively resolving faults for thousands of subscribers
- Performed backend PSTN and VoIP configuration via Nokia, Huawei, ZTE and Interaction network management portals; created and managed trunking groups using IBM BPM
- Managed the full incident lifecycle in Oracle CRM — logging, prioritisation, SLA tracking, escalation and resolution — and supported field engineers on complex on-site troubleshooting

KEY PROJECTS

Enterprise MES/ERP Platform — Sage Evolution Integrated

Jul 2025 — Present

Kingsley Beverage FZCO | *Live demo: saadm.dev/app/*

- Sole developer of a 20+ module full-stack platform: production planning, inventory, QC, procurement, dispatch, accounts and live production reporting
- Stack: Python (FastAPI, async), MongoDB, SQL Server, Docker Compose (API + MongoDB + Nginx + TLS), JavaScript, Tailwind CSS, fpdf2 for automated document generation, deployed on a self-managed Linux VM behind Cloudflare

Sanad — AI Customer Support Copilot

Oct 2025 — Present

Personal project | *Live demo: saadm.dev/sanad/*

- Production SaaS helpdesk with Claude AI across every workflow touchpoint: RAG knowledge-base retrieval with citations, streaming responses, EN/AR translation, sentiment classification and automated ticket routing
- Admin console with live model switching (Haiku / Sonnet / Opus); server-side API key management via a Cloudflare Worker proxy with rate limiting and streaming passthrough

ShopFloor API — MES / OEE Backend

2026

Open source | *Java 21 · Spring Boot 3*

- Production-floor operations REST API: job orders computing OEE (Availability × Performance × Quality) on close, downtime with root-cause logging, QC holds and FIFO inventory across stock lots
- Spring Security with JWT role enforcement, JPA + PostgreSQL + Flyway migrations, JUnit 5 / MockMvc / Testcontainers integration tests with green CI

hft-orderbook — Low-Latency Trading Engine

2026

Open source | *C++17 · NASDAQ ITCH 5.0 · FIX 4.4*

- Reconstructs limit-order books from the real NASDAQ TotalView-ITCH 5.0 feed at well under 100 ns/message via a wait-free SPSC ring; symbols shard across pinnable worker threads, validated race-free with ThreadSanitizer
- Microstructure signals (micro-price, order-book imbalance, spread in bps), FIX 4.4 order entry and a MetaTrader 5 bridge; hardened with Address/UB sanitizers and a libFuzzer decode harness in CI

Automated Network Configuration Tool

Jul 2023 — Jul 2025

PTCL | *Rawalpindi, Pakistan*

- Python automation system driven by a structured database of area codes, number ranges and network parameters, auto-generating ready-to-use configuration scripts from incoming service tickets

TECHNICAL SKILLS

Languages: Python (advanced), JavaScript, TypeScript, Java, C/C++, SQL, MATLAB

Backend & Frameworks: FastAPI (async), Django, Spring Boot, REST API design, JWT auth, microservices

AI & LLM: Claude API (Anthropic), OpenAI API, LangChain, Retrieval-Augmented Generation (RAG), prompt engineering, streaming responses, sentiment classification

Frontend: React, TypeScript, JavaScript, Tailwind CSS, HTML/CSS

Data: MongoDB, Microsoft SQL Server, PostgreSQL, Sage Evolution ERP integration, data analysis

Infrastructure & DevOps: Docker & Docker Compose, Linux, Nginx, Cloudflare, VMware, NAS/RAID, Dell server administration, Git, GitHub Actions CI

MES & ERP: Full-stack MES/ERP development, production planning (PPC), inventory management, OEE monitoring, batch tracking, FIFO/FEFO traceability, automated PDF reporting

Industrial & Networking: Krones production lines (blow molding, filling, Checkmate, Variopac FS, palletizing, PET preform), electrical troubleshooting, RCA, preventive maintenance, GPON, PSTN, broadband infrastructure, Oracle CRM, SolarWinds, IBM BPM

Languages spoken: English — Fluent (IELTS certified) | Urdu — Native

CERTIFICATIONS

- RAG and Agentic AI — IBM

- Generative AI Automation — IBM
- AI Engineering — IBM

EDUCATION

B.Sc. Electrical Engineering — Computer Engineering Major Sep 2020 – Sep 2024
COMSATS University Islamabad | Pakistan · GPA 3.0
Relevant coursework: Python, C++, MATLAB, computer networks, operating systems, embedded systems, databases, algorithms, software engineering

F.Sc. Pre-Engineering 2018 – 2020
Fazaia Intermediate College | Rawalpindi, Pakistan · 75%

INTERNSHIPS

Machine Learning Intern 2023
Omdena | Remote

- Built machine learning models for accessibility applications and environmental monitoring as part of a global collaborative AI project, covering model training, evaluation and deployment pipelines

Technical Engineering Intern Jun 2023 – Aug 2023
Kingsley Beverage FZCO | Dubai, UAE · On-site

- Hands-on training on automated Krones lines; developed and maintained SOPs, work instructions and quality manuals; troubleshoot machine faults under supervision

Network Engineering Intern Jun 2022 – Aug 2022
PTCL | Rawalpindi, Pakistan · On-site

- Data centre migration: rack and stack, structured cabling, fibre splicing, patch panel termination and Cisco routing/switching configuration; monitored network performance and supported fault resolution